STAT 8561: Linear Statistical Analysis 1

Homework 2

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Question 1: Least Squares: Assumptions and Consequences [6]

Consider $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is fixed, has full column rank, and $n > p$. Select one answer per part.

(a) Changing the distributional assumption [2]

Two students use the same observed $\mathbf{X}$ and $\mathbf{Y}$. One assumes normal errors; the other does not specify an error distribution. Both define their estimate by minimizing

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Which statement is correct?

A. The second student cannot compute a least squares estimate.
B. The estimates must differ because the distributional assumptions differ.
C. The estimates are the same because they minimize the same criterion for the same data.
D. The estimates are the same only if the observed residuals are all zero.

(b) What makes the minimizer unique? [2]

Which statement explains why solving the normal equations gives a unique least squares minimizer under the stated assumptions?

A. Full column rank makes $\mathbf{X}^\top\mathbf{X}$ positive definite.
B. Normal errors make every system of normal equations invertible.
C. Having more observations than parameters guarantees full column rank for any design matrix.
D. Minimizing the sum of squared residuals forces every residual to be zero.

(c) Shifting every response [2]

Now suppose the model includes an intercept. Add 5 to every observed response, keep all predictors unchanged, and refit the same model by least squares. What happens?

A. Every slope increases by 5, while the intercept is unchanged.
B. The coefficients are unchanged, and every residual increases by 5.
C. The intercept increases by 5, and every residual increases by 5.
D. The intercept increases by 5, the slopes are unchanged, and the residuals are unchanged.

Ans:

a.) C

Least squares is just an optimization problem on the data: we are minimizing $S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$. Since this criterion depends only on $\mathbf{X}$ and $\mathbf{Y}$ and doesn't rely on any distribution for the error term $\boldsymbol{\varepsilon}$, both students are solving the exact same minimization problem,

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y},$$

and arrive at the exact same estimate. The distributional assumptions (like normality) only matter later when doing inference and confidence intervals, not for finding the point estimate itself.

b.) A

Taking the gradient of $S(\boldsymbol{\beta})$, we have $\nabla S(\boldsymbol{\beta}) = -2\mathbf{X}^\top\mathbf{Y} + 2\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}$. Differentiating once more gives the Hessian,

$$\frac{\partial^2 S(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}\,\partial\boldsymbol{\beta}^\top} = 2\mathbf{X}^\top\mathbf{X}.$$

For any non-zero vector $\mathbf{v} \neq \mathbf{0}$,

$$\mathbf{v}^\top\mathbf{X}^\top\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 > 0,$$

which is strictly positive because $\mathbf{X}$ having full column rank means $\mathbf{X}\mathbf{v} = \mathbf{0}$ only if $\mathbf{v} = \mathbf{0}$. Thus, $\mathbf{X}^\top\mathbf{X}$ is positive definite, which means it is invertible and $S(\boldsymbol{\beta})$ is strictly convex, giving a unique minimizer. Option C fails because simply having $n > p$ does not guarantee that the columns of $\mathbf{X}$ are linearly independent.

c.) D

Let the new response vector be $\mathbf{Y}^* = \mathbf{Y} + 5 \cdot \mathbf{1}_n$. Because the model includes an intercept, $\mathbf{1}_n$ is the first column of $\mathbf{X}$, which we can write as $\mathbf{1}_n = \mathbf{X}\mathbf{u}_1$ where $\mathbf{u}_1 = (1, 0, \ldots, 0)^\top$. Refitting gives,

$$\begin{aligned}
\hat{\boldsymbol{\beta}}^* &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{Y} + 5\mathbf{X}\mathbf{u}_1) \\
&= \hat{\boldsymbol{\beta}} + 5(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\mathbf{u}_1 \\
&= \hat{\boldsymbol{\beta}} + 5\mathbf{u}_1
\end{aligned}$$

So only the intercept increases by 5, while all the slopes stay the same. For the residuals,

$$\mathbf{e}^* = \mathbf{M}\mathbf{Y}^* = \mathbf{M}\mathbf{Y} + 5\mathbf{M}\mathbf{1}_n = \mathbf{e} + \mathbf{0} = \mathbf{e}$$

since $\mathbf{1}_n$ is in the column space of $\mathbf{X}$ and $\mathbf{M}$ projects onto the orthogonal complement, annihilating $\mathbf{1}_n$. The residuals do not change at all.

□

Question 2: Projection Geometry and Rank [8]

For parts (a) and (c), let $\mathbf{X}$ have full column rank. Define

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \quad \hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{e} = \mathbf{M}\mathbf{Y}.$$

(a) Projecting a second time [2]

A student applies $\mathbf{H}$ to the fitted vector and to the residual vector. Which pair of results must hold?

A. $\mathbf{H}\hat{\mathbf{Y}} = \mathbf{0}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.
B. $\mathbf{H}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}$ and $\mathbf{H}\mathbf{e} = \mathbf{0}$.
C. $\mathbf{H}\hat{\mathbf{Y}} = \mathbf{Y}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.
D. $\mathbf{H}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

(b) Adding a redundant predictor [2]

Suppose $\mathbf{x}$ is not constant. Compare the least squares fits using

$$\mathbf{X} = [\mathbf{1}_n, \mathbf{x}] \quad \text{and} \quad \mathbf{W} = [\mathbf{1}_n, \mathbf{x}, 2\mathbf{x}]$$

for the same response vector. Which statement is correct?

A. The extra column must strictly reduce SSE.
B. The fit using $\mathbf{W}$ does not exist because $\mathbf{W}^\top\mathbf{W}$ is singular.
C. The fit using $\mathbf{W}$ has nonunique coefficients, but its fitted values and SSE equal those from $\mathbf{X}$.
D. The fit using $\mathbf{W}$ has nonunique coefficients and therefore nonunique fitted values.

(c) Geometric consequences [4]

Enter T or F. Each statement is worth 1 point. Here $\bar{Y} = n^{-1}\sum_i Y_i$.

Statement	T/F
(i) The uncentered identity $\mathbf{Y}^\top\mathbf{Y} = \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e}$ holds for the least squares fit.	____
(ii) The centered identity $\sum_i (Y_i - \bar{Y})^2 = \sum_i (\hat{Y}_i - \bar{Y})^2 + \text{SSE}$ is guaranteed for every least squares model, even one without an intercept.	____
(iii) If the model includes an intercept, the average fitted value equals the average observed response.	____
(iv) The residual vector must be orthogonal to the observed response vector $\mathbf{Y}$ itself.	____

Ans:

a.) B

$\mathbf{H}$ is an orthogonal projection matrix, so it is idempotent:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{H}$$

and $\mathbf{H}\mathbf{M} = \mathbf{H}(\mathbf{I}_n - \mathbf{H}) = \mathbf{H} - \mathbf{H}^2 = \mathbf{0}$. Therefore,

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}\mathbf{H}\mathbf{Y} = \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}$$
$$\mathbf{H}\mathbf{e} = \mathbf{H}\mathbf{M}\mathbf{Y} = \mathbf{0}$$

Geometrically, $\hat{\mathbf{Y}}$ is already in the column space $\mathcal{C}(\mathbf{X})$, so projecting it onto that space again does not change it. On the other hand, the residual $\mathbf{e}$ is orthogonal to $\mathcal{C}(\mathbf{X})$, so projecting it onto $\mathcal{C}(\mathbf{X})$ drops it straight to $\mathbf{0}$.

b.) C

Because the third column $2\mathbf{x}$ is a scalar multiple of $\mathbf{x}$, $\text{rank}(\mathbf{W}) = 2 < 3$, meaning $\mathbf{W}^\top\mathbf{W}$ is singular and cannot be inverted. This gives infinitely many choices for the coefficient vector $\boldsymbol{\beta}$.

However, adding a redundant column does not change the span:

$$\mathcal{C}(\mathbf{W}) = \text{span}\{\mathbf{1}_n, \mathbf{x}, 2\mathbf{x}\} = \text{span}\{\mathbf{1}_n, \mathbf{x}\} = \mathcal{C}(\mathbf{X})$$

Since $\hat{\mathbf{Y}}$ is simply the orthogonal projection of $\mathbf{Y}$ onto this column space, and projection onto a subspace is uniquely defined, the fitted values $\hat{\mathbf{Y}}$, the residuals $\mathbf{e}$, and $\text{SSE} = \mathbf{e}^\top\mathbf{e}$ are exactly the same as in the original model. So a fit definitely exists (making B false), but the coefficients are not unique.

c.) i.) True

Writing $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$, we can expand the sum of squares:

$$\begin{aligned}\mathbf{Y}^\top\mathbf{Y} &= (\hat{\mathbf{Y}} + \mathbf{e})^\top(\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + 2\hat{\mathbf{Y}}^\top\mathbf{e} + \mathbf{e}^\top\mathbf{e} \\ &= \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e}\end{aligned}$$

The cross-term is $\hat{\mathbf{Y}}^\top\mathbf{e} = \mathbf{Y}^\top\mathbf{H}\mathbf{M}\mathbf{Y} = 0$ since $\mathbf{H}\mathbf{M} = \mathbf{0}$. This is just the Pythagorean theorem for orthogonal vectors, and it holds whether or not there is an intercept in the model.

ii.) False

If we center by subtracting $\bar{Y}\mathbf{1}_n$, we get:

$$\sum_i (Y_i - \bar{Y})^2 = \sum_i (\hat{Y}_i - \bar{Y})^2 + \text{SSE} + 2\sum_i (\hat{Y}_i - \bar{Y})e_i$$

The cross-term is $2\hat{\mathbf{Y}}^\top\mathbf{e} - 2\bar{Y}\sum_i e_i = -2\bar{Y}\sum_i e_i$. This only drops to zero if $\sum_i e_i = 0$, which is only guaranteed when $\mathbf{1}_n \in \mathcal{C}(\mathbf{X})$ (meaning the model has an intercept). Without an intercept, $\sum_i e_i \neq 0$ in general, so the identity fails.

iii.) True

When the model includes an intercept, $\mathbf{1}_n$ is in the column space of $\mathbf{X}$, so $\mathbf{1}_n^\top\mathbf{e} = 0$, which means $\sum_i e_i = 0$. Then,

$$\sum_i Y_i = \sum_i \hat{Y}_i + \sum_i e_i = \sum_i \hat{Y}_i \implies \frac{1}{n}\sum_i \hat{Y}_i = \frac{1}{n}\sum_i Y_i$$

so the average fitted value equals the average observed response.

iv.) False

Multiplying $\mathbf{e}^\top$ by $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$ gives:

$$\mathbf{e}^\top\mathbf{Y} = \mathbf{e}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e} = 0 + \text{SSE} = \text{SSE}$$

This is only zero in the trivial case where the fit is perfect ($\text{SSE} = 0$). Residuals are orthogonal to the fitted values $\hat{\mathbf{Y}}$ and the predictors in $\mathbf{X}$, but they are not orthogonal to $\mathbf{Y}$ itself.

□

Question 3: Sampling Properties and the Role of Normality [8]

Throughout this question, $\mathbf{X}$ is fixed with full column rank, $n > p$, and $\sigma^2 > 0$.

(a) Match the result to its assumptions [4]

Choose the weakest sufficient set of assumptions among A, B, and C for each result. Each row is worth 1 point. Letters may be reused.

A. The least squares criterion and full column rank alone; no error moment or distribution assumptions.
B. $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$, without assuming normality.
C. The normal linear model, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I}_n)$.

Result	Letter
(i) The least squares coefficient vector is the unique minimizer of $S(\boldsymbol{\beta})$.	____
(ii) $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$.	____
(iii) $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$.	____
(iv) $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$.	____

(b) Why does independence follow? [2]

Under the normal linear model, which argument correctly explains why $\hat{\boldsymbol{\beta}}$ and SSE are independent?

A. $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are jointly normal with zero covariance, and SSE is a function of $\mathbf{e}$.
B. Any two uncorrelated random vectors are independent, regardless of their distribution.
C. They estimate different parameters, so they must be independent.
D. Each observed residual is independent of every other observed residual in every normal linear model.

(c) Increasing the error variance [2]

Keep $\mathbf{X}$ and $\boldsymbol{\beta}$ fixed. In a second model, the errors still have mean zero, but their covariance is $4\sigma^2\mathbf{I}_n$ instead of $\sigma^2\mathbf{I}_n$. Which statement describes the sampling distribution's mean and covariance for the OLS estimator?

A. Its mean and covariance are both multiplied by 4.
B. Its mean is unchanged, and its covariance is multiplied by 16.
C. Its mean and covariance are both unchanged.
D. Its mean is unchanged, and its covariance is multiplied by 4; each coefficient's standard deviation doubles.

Ans:

a.) i.) A

Minimizing $S(\boldsymbol{\beta})$ is purely an algebraic problem on fixed data. Having full column rank makes $\mathbf{X}^\top\mathbf{X}$ invertible and $S(\boldsymbol{\beta})$ strictly convex, guaranteeing a unique minimizer without assuming anything about the errors $\boldsymbol{\varepsilon}$.

ii.) B

Writing out the estimator,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$$

Taking the expected value gives,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta}$$

This requires the errors to have mean zero ($\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$), so A is not enough. But assumption B provides this without needing full normality, making B the weakest sufficient set.

iii.) C

We can write $\text{SSE}/\sigma^2 = \boldsymbol{\varepsilon}^\top \mathbf{M}\boldsymbol{\varepsilon}/\sigma^2$. This is a quadratic form with $\mathbf{M}$ symmetric idempotent and $\text{rank}(\mathbf{M}) = n - p$. To get the exact χ^2_{n-p} distribution, the errors must follow a normal distribution. Under B alone, we can find the expected value $\mathbb{E}[\text{SSE}] = \sigma^2(n-p)$, but we cannot claim it follows a chi-square distribution without normality.

iv.) B

Using the variance formula,

$$\begin{aligned}\text{Var}(\hat{\boldsymbol{\beta}}) &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \text{Var}(\boldsymbol{\varepsilon})\, \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\end{aligned}$$

This only requires the covariance of the errors to be $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$. We don't need normality to find the variance of the estimator, so B is sufficient.

b.) A

Both $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are linear transformations of the error vector $\boldsymbol{\varepsilon}$:

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}, \qquad \mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$$

Under the normal linear model, they are jointly normal. Computing their covariance gives,

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\sigma^2\mathbf{I}_n)\mathbf{M}^\top = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{M}\mathbf{X})^\top = \mathbf{0}$$

because $\mathbf{M}\mathbf{X} = \mathbf{X} - \mathbf{H}\mathbf{X} = \mathbf{0}$. For jointly normal variables, having zero covariance implies independence. Since $\text{SSE} = \mathbf{e}^\top\mathbf{e}$ is just a function of the residual vector $\mathbf{e}$, it must be independent of $\hat{\boldsymbol{\beta}}$. Option B is false because zero covariance only implies independence under normality, and D is false because the covariance matrix of $\mathbf{e}$ is $\sigma^2\mathbf{M}$, which is not diagonal.

c.) D

The expectation of $\hat{\boldsymbol{\beta}}$ only depends on the mean of the errors, which is still zero, so $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ is unchanged.

Replacing σ^2 with $4\sigma^2$ in the covariance formula gives,

$$\text{Var}(\hat{\boldsymbol{\beta}}) = 4\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$$

So the covariance matrix is multiplied by 4 (not 16, since the variance factor enters linearly). Taking the square root for the standard errors, each coefficient's standard deviation is multiplied by $\sqrt{4} = 2$, meaning it doubles.

□

Question 4: Interpreting Confidence and Prediction Intervals [6]

Assume a correctly specified normal linear model. For prediction, a future observation has an error independent of the original data and with the same error variance.

(a) Mean response versus one future response [2]

At the same predictor value, software reports two intervals at the 95% level:

- Interval I: (2.493, 3.507);
- Interval II: (1.659, 4.341).

One is a confidence interval for the mean response; the other is a prediction interval for one future response. Which interpretation is correct?

A. I is the prediction interval because predicting one observation involves less uncertainty than estimating a mean.
B. Both intervals describe the same target; their different widths are only rounding error.
C. I is the mean-response interval; II is the prediction interval because it also accounts for the new observation's error.
D. II is the mean-response interval because averaging adds the new observation's error variance.

(b) Connecting a confidence interval and a test [2]

A 95% confidence interval for β_1 is $(0.4,\ 1.6)$. Consider the corresponding two-sided t test of

$$H_0 : \beta_1 = 1 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 1$$

at $\alpha = 0.05$. Which conclusion is correct?

A. Reject H_0 because zero is outside the interval.
B. Fail to reject H_0 because 1 is inside the interval; this does not prove that $\beta_1 = 1$.
C. Accept H_0 as proven true because 1 is inside the interval.
D. No conclusion is possible because a t test can only test a coefficient against zero.

(c) What an interval means [2]

Enter T or F. Each statement is worth 1 point. Assume a positive estimated error variance and $\mathbf{x}_0^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_0 > 0$.

Statement	T/F
(i) Holding the data and predictor value fixed, changing a mean-response confidence level from 95% to 99% makes the interval wider.	____
(ii) A 95% confidence interval for the mean response is designed to contain 95% of individual future responses at that predictor value.	____

Ans:

a.) C

At a given predictor value $\mathbf{x}_0$, the formulas for the two intervals are:

$$\text{CI:} \quad \hat{Y}_0 \pm t_{0.975,n-p}\,\hat{\sigma}\sqrt{\mathbf{x}_0^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_0}$$

$$\text{PI:} \quad \hat{Y}_0 \pm t_{0.975,n-p}\,\hat{\sigma}\sqrt{1+\mathbf{x}_0^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_0}$$

The prediction interval has that extra $+1$ inside the square root to account for the random error ε_0 of the new observation, so it is always wider than the confidence interval for the mean response.

Looking at the numbers, both intervals are centered at $\hat{Y}_0 = 3.0$. Interval I has half-width $3.507 - 3 = 0.507$, while Interval II has half-width $4.341 - 3 = 1.341$. Since Interval II is wider, it must be the prediction interval and Interval I is the mean-response confidence interval.

b.) B

For a two-sided level-α test of $H_0 : \beta_1 = c$, we reject the null hypothesis if and only if c falls outside the $100(1-\alpha)\%$ confidence interval.

Here we are testing $c = 1$, and 1 lies comfortably inside the interval $(0.4, 1.6)$. Therefore, we fail to reject H_0. Failing to reject doesn't mean we have proven that $\beta_1 = 1$; it just means 1 is plausible given the data (as

are all other values in the interval). Option A is wrong because zero being outside the interval would test whether $\beta_1 = 0$, not $\beta_1 = 1$.

c.) i.) True

The half-width of the confidence interval is $t_{1-\alpha/2,n-p}\,\mathrm{SE}(\hat{Y}_0)$. Increasing the confidence level from 95% to 99% increases the critical value ($t_{0.995,n-p} > t_{0.975,n-p}$). Since the standard error is held fixed, the interval has to become wider to have higher confidence.

ii.) False

A confidence interval for the mean response is designed to estimate the mean $\mathbb{E}[Y_0] = \mathbf{x}_0^\top \boldsymbol{\beta}$, not individual responses. Individual observations scatter around the mean with variance σ^2, so containing future observations requires a prediction interval, not a mean-response confidence interval.

□

Question 5: Distribution and Inference [12]

Suppose a multiple linear regression model has $n = 20$ observations and $p = 3$ parameters: an intercept and two slopes. The first column of $\mathbf{X}$ is $\mathbf{1}_{20}$. Assume

$$\mathbf{Y} \sim N_{20}\big(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_{20}\big)\,.$$

Suppose

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} 2.0 \\ 1.5 \\ -0.5 \end{pmatrix}, \qquad (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} 0.05 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.08 \end{pmatrix},$$

and $\mathrm{SSE} = 34$. Use $t_{0.975,17} = 2.110$.

(a) [3]

Choose the correct distributions.

For $\hat{\boldsymbol{\beta}}$:

A. $N_3\big(\mathbf{0}, \sigma^2 \mathbf{I}_3\big)$
B. $N_3\big(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\big)$
C. t_{17}
D. χ^2_{17}

For SSE/σ^2:

A. $N(0,1)$
B. t_{17}
C. χ^2_{17}
D. $F_{3,17}$

(b) [2]

Compute

$$\hat{\sigma}^2 = \frac{\mathrm{SSE}}{n-p} = _____.$$

(c) [4]

For testing

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 0,$$

where β_1 is the second entry of $\boldsymbol{\beta}$, fill in the following. Use significance level $\alpha = 0.05$.

$$\mathrm{SE}(\hat{\beta}_1) = _____, \qquad t = _____,$$

and choose the decision:

A. Reject H_0
B. Fail to reject H_0

(d) [3]

Which of the following is the 95% confidence interval for β_1?

A. $(0.8329,\ 2.1671)$
B. $(1.1838,\ 1.8162)$
C. $(-0.6671,\ 0.6671)$
D. $(0,\ 3.0000)$

Ans:

a.) $\hat{\boldsymbol{\beta}}$: B; SSE/σ^2: C

Since $\mathbf{Y}$ is normally distributed, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ is a linear combination of normal random variables, so it is also normally distributed:

$$\hat{\boldsymbol{\beta}} \sim N_3\big(\boldsymbol{\beta},\, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\big)$$

For the sum of squared errors, $\mathrm{SSE}/\sigma^2 = \boldsymbol{\varepsilon}^\top\mathbf{M}\boldsymbol{\varepsilon}/\sigma^2$ follows a chi-square distribution with degrees of freedom equal to $\mathrm{rank}(\mathbf{M}) = n - p = 20 - 3 = 17$:

$$\frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_{17}$$

b.) $\hat{\sigma}^2 = 2.000$

Computing the estimate for σ^2,

$$\hat{\sigma}^2 = \frac{\mathrm{SSE}}{n-p} = \frac{34}{20-3} = \frac{34}{17} = 2.000$$

c.) $\mathrm{SE}(\hat{\beta}_1) = 0.316,\ \ t = 4.743,\ \ \text{decision: A}$

The parameter β_1 is the second element of $\boldsymbol{\beta}$, so its variance uses the $(2,2)$ diagonal entry of $(\mathbf{X}^\top\mathbf{X})^{-1}$:

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{\hat{\sigma}^2\,[(\mathbf{X}^\top\mathbf{X})^{-1}]_{22}} = \sqrt{2(0.05)} = \sqrt{0.1} \approx 0.31623$$

Rounding to three decimal places gives $\mathrm{SE}(\hat{\beta}_1) = 0.316$.

Under the null hypothesis $H_0 : \beta_1 = 0$, the test statistic is,

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)} = \frac{1.5}{0.31623} \approx 4.74342$$

Rounding to three decimal places gives $t = 4.743$.

Comparing with the critical value, $|t| = 4.743 > t_{0.975,17} = 2.110$, so we reject H_0 (Option A).

d.) A

Using the formula for the 95% confidence interval,

$$\begin{aligned}\hat{\beta}_1 \pm t_{0.975,17}\,\mathrm{SE}(\hat{\beta}_1) &= 1.5 \pm 2.110(0.31623)\\ &= 1.5 \pm 0.66724\\ &= (0.8328,\ 2.1672)\end{aligned}$$

This matches option A: $(0.8329, 2.1671)$ (the minor difference in the last digit is just rounding). Option B forgets the t multiplier, and C is centered around 0 instead of $\hat{\beta}_1$. Since 0 is not in the confidence interval, this also agrees with rejecting H_0 in part (c).

□